

How does JavaScript code run behind the scenes?(for "var" keyword)

When we run a code in JavaScript, then "Execution Context" is created.

Means, Whenever a JS code is executed/runs, an execution context is created for that code which contains two phases or which means code runs in two phases:

1. Memory Allocation Phase
2. Code Execution Phase

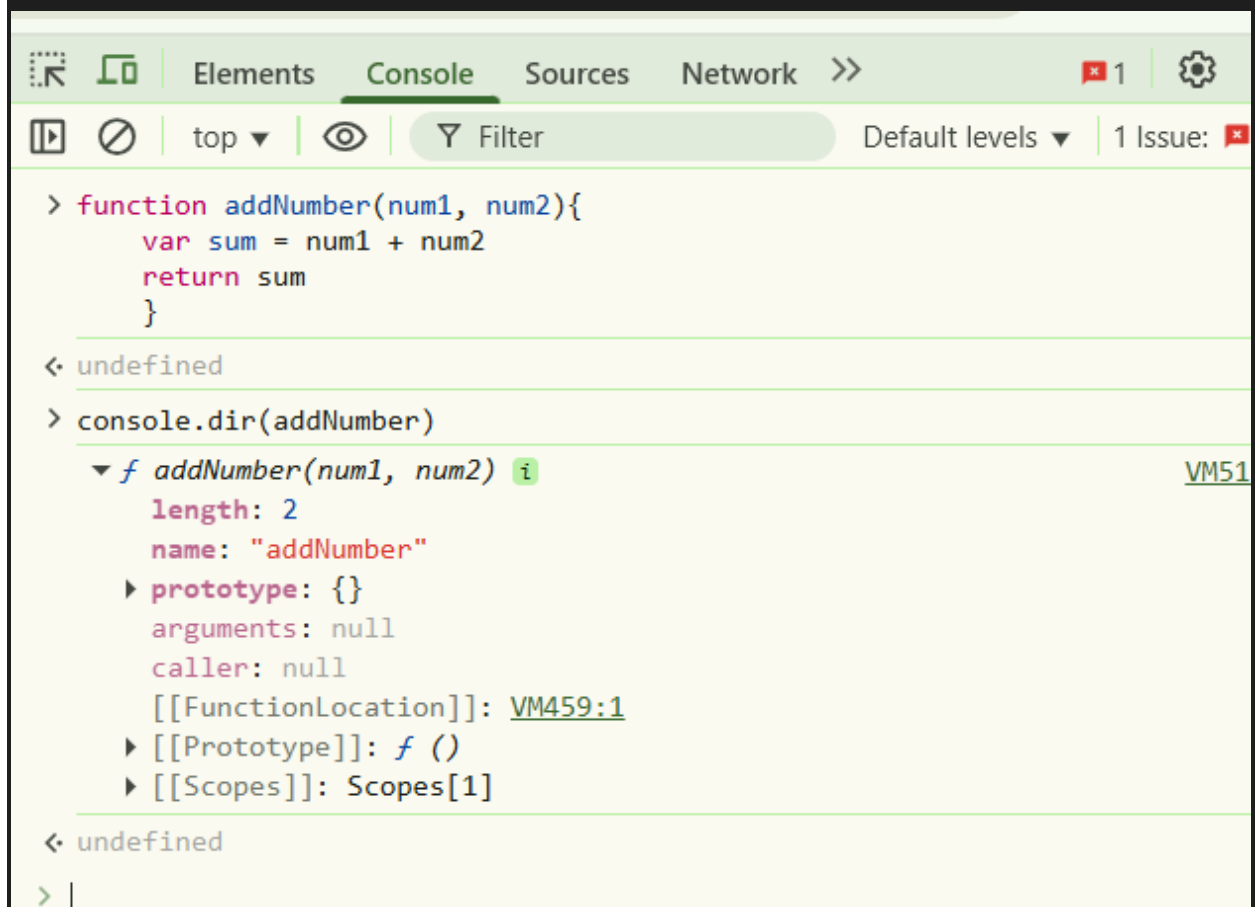
1. Memory Allocation Phase:

In this phase, only memory is allocated for variables and functions.

Variables are initialized with 'undefined', and functions are stored in memory with their name, defination/code/declaration & no. of parameters(if any) in the heap.

And, for a function initially there is no memory allocation happens for code(variable etc.) written inside that function. Because, initailly when memory allocation happens for a function, then code inside the function does not come in the execution context. We can check this by writing a function & then, "console.dir(fn_name)" in the browser's

console. Below image shows this...



Memory allocation for code written inside functions will happen when the **function** is invoked or called.

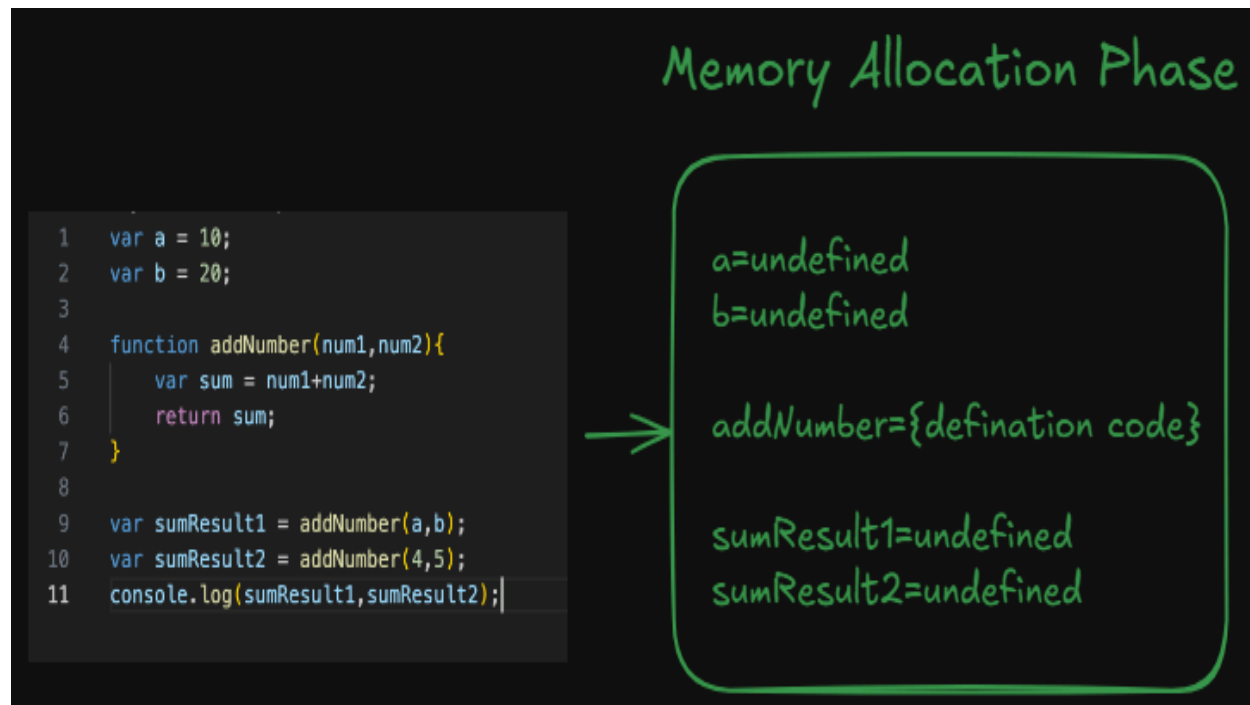
OR

Code inside the function will be stored in memory only when the function is called/invoked.

OR

Code inside the function comes in execution context only when the function is invoked/called.

Memory Allocation Phase For A Code



2. Code Execution Phase:

In **this** phase, the code is executed line by line. Variables are assigned to their actual values, and functions are executed when called.

In this phase, if a **function** is invoked/called, then code inside the function comes into the execution context's memory allocation phase where memory allocation happens for code (variables /parameters etc. exist) inside the function.

After that, code inside the **function** comes in the execution phase where variables(parameters) are assigned to their actual values & rest of the code is executed.

Execution Phase For A Code

